

HOMS

# Mini-Assessment Memorandum

## Motion in One Dimension

Physical Sciences | Grade 10 | Term 3

Marking guidance and worked answers. Total: 30 marks.

|              |  |
|--------------|--|
| Learner name |  |
| Date         |  |

*CAPS-aligned draft. Educator or subject-expert approval is required before classroom use.*

**Marking principle:** Credit equivalent correct methods, units and directions. Do not award a final-answer mark when contradictory working is shown.

## Question 1

- 1.1 Displacement divided by the time interval. [2]
- 1.2 A vector has magnitude and direction; a scalar has magnitude only. [2]
- 1.3 No. [2]
- At an instant an object may have zero velocity while its velocity is changing, so acceleration may be non-zero.
- 1.4 The position is constant and the object is at rest. [2]

## Question 2

- 2.1 Distance = 160 m; average speed =  $160/20 = 8.0 \text{ m.s}^{-1}$ . [3]
- 2.2 Displacement = +80 m; average velocity =  $80/20 = 4.0 \text{ m.s}^{-1}$  east. [3]
- 2.3  $v = 6 + (1.5)(8) = 18 \text{ m.s}^{-1}$ . [2]
- 2.4  $\Delta x = (6)(8) + 0.5(1.5)(8^2) = 48 + 48 = 96 \text{ m}$ . [4]

## Question 3

- 3.1  $a = (8 - 0)/(4 - 0) = 2.0 \text{ m.s}^{-2}$ . [2]
- 3.2 The object moves at constant positive velocity of  $8 \text{ m.s}^{-1}$ . [2]
- 3.3  $a = (0 - 8)/(12 - 8) = -2.0 \text{ m.s}^{-2}$ . [2]
- 3.4 Area =  $0.5(4)(8) + (4)(8) + 0.5(4)(8) = 16 + 32 + 16 = 64 \text{ m}$ . [4]

**TOTAL: 30**